C1 – Outcomes, Support and PMFs

Global Economics | Grade 10 | Term 2 | Lesson 1

C1: Characterises discrete random variables in economic and financial contexts by identifying their support, probability mass function and possible outcomes.

A random variable assigns a number to each complete outcome. Its probability mass function (PMF) gives the probability of each value. The support contains the values with positive probability.

Worked examples

1. Changes to the till balance — Type A

A shop sells a jacket for 12 USD. Exactly one of the events in the table happens by the end of the return period.

Task 1. Identify the outcomes and their probabilities. Read the four possible outcomes and their probabilities from the table.

Complete outcome	Kept	Even exchange	Upgrade	Full refund
Outcome probability	0.37	0.21	0.16	0.26

Task 2. Identify the random variable values and their probabilities. Map the outcomes and group the values. Let X be the change in the shop's till balance after the original payment, in USD. Keeping the jacket or exchanging it at the same price gives $X = 0$. The upgrade gives $X = 3$ and the full refund gives $X = -12$. Add the probabilities of the two outcomes that give $X = 0$.

x (USD)	-12	0	3
$p_X(x) = P(X = x)$	0.26	$0.37 + 0.21 = 0.58$	0.16

The support is $\{-12, 0, 3\}$ in USD. All masses are nonnegative and $0.26 + 0.58 + 0.16 = 1$.

Task 3. Find a specific probability. Find the probability that the balance stays the same after the original payment. The PMF gives $P(X = 0) = p_X(0) = 0.58$. The balance stays the same in 58% of these sales.

2. Invoice payment adjustment — Type B

Two customers independently pay their invoices by the deadline (T) with probabilities 0.70 and 0.55, respectively, or pay late (L). The business earns an 8 USD bonus if both pay by the deadline, pays a 6 USD charge if both pay late, and has no adjustment otherwise.

Task 1. Identify the outcomes and their probabilities. List every pair in customer order and calculate its probability. Late-payment probabilities are $1 - 0.70 = 0.30$ and $1 - 0.55 = 0.45$. Independence lets us multiply within each pair.

Ordered outcome	TT	TL	LT	LL
Outcome probability	$0.70 \cdot 0.55 = 0.385$	$0.70 \cdot 0.45 = 0.315$	$0.30 \cdot 0.55 = 0.165$	$0.30 \cdot 0.45 = 0.135$

Task 2. Identify the random variable values and their probabilities. Map the outcomes and build the PMF. Let Y be the payment adjustment in USD. The pairs TT, TL, LT, and LL give $8, 0, 0, -6$, respectively. Add the probabilities of the two pairs that give $Y = 0$.

y (USD)	-6	0	8
$p_Y(y) = P(Y = y)$	0.135	$0.315 + 0.165 = 0.480$	0.385

The support is $\{-6, 0, 8\}$ in USD. All masses are nonnegative and $0.135 + 0.480 + 0.385 = 1$.

Task 3. Find a specific probability. Find the probability of no payment adjustment. The PMF gives $P(Y = 0) = p_Y(0) = 0.480$. The business has no adjustment in 48% of these invoice pairs.

Guided practice

3. Courier pay adjustment — Type A

Each delivery falls into exactly one category below. An early or on-time delivery adds 2 USD to the courier's base pay, a slightly late delivery leaves it unchanged, and a very late delivery deducts 3 USD.

Task 1. Identify the outcomes and their probabilities. Read the four delivery outcomes and their supplied probabilities in the first two rows.

Complete outcome	Early	On time	Slightly late	Very late
Outcome probability	0.18	0.47	0.23	0.12
A (USD)	2			

Task 2. Identify the random variable values and their probabilities. Define the pay adjustment A, with units. Complete its mapping row above. Group outcomes by value to build the PMF below. Give the support and verify nonnegative masses and a sum of 1.

a (USD)	Outcomes giving a	Calculation and mass $p_A(a)$
-3		
0		
2		

Task 3. Find a specific probability. Use the PMF to find $P(A = 2)$, the probability that the courier earns exactly 2 USD above base pay. Interpret the result.

4. Delivery fees from two orders — Type B

Two customers independently keep (K) or cancel (C) their orders. Their cancellation probabilities are 0.20 and 0.35, respectively. The shop earns 3 USD per kept order and no fee per cancellation.

Task 1. Identify the outcomes and their probabilities. Complete the first two rows. List every keep/cancel outcome in customer order and multiply within each pair to calculate its probability.

Ordered outcome	KK			
Outcome probability	$0.80 \cdot 0.65 = 0.52$			
F (USD)				

Task 2. Identify the random variable values and their probabilities. Define the total fee revenue F, with units. Complete its mapping row above. Group outcomes by value to build the PMF below. Give the support and verify nonnegative masses and a sum of 1.

f (USD)	Outcomes giving f	Calculation and mass $p_F(f)$
0		
3		
6		

Task 3. Find a specific probability. Use the PMF to find $P(F = 3)$, the probability that the shop earns exactly 3 USD in delivery fees. Interpret the result.

Independent practice

5. Buying a carton of fruit **Type A**

A food stall buys one carton of fruit. Exactly one supplier fills the order. The probabilities for the orchard, cooperative, wholesaler, and grocery are 0.28, 0.19, 0.31, and 0.22, respectively. Their prices for the carton are 14, 18, 14, and 22 USD, in the same order.

Task 1. Identify the outcomes and their probabilities. Record every supplier outcome and its given probability.

Task 2. Identify the random variable values and their probabilities. Define C, the purchase cost, with units. Map every outcome to its value of C. List the support. Group outcomes by value to build the PMF, show your calculations, and verify nonnegative masses and a sum of 1.

Task 3. Find a specific probability. Use the PMF to find $P(C = 14)$, the probability that the stall pays exactly 14 USD for the carton. Interpret the result.

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6. Demand and a supplier surcharge **Type B**

On a market day, a shop has high demand (H) with probability 0.30 and regular demand (R) otherwise. Independently, its supplier charges a surcharge (S) with probability 0.25 and charges no surcharge (N) otherwise. High demand adds 6 USD to profit relative to the shop's baseline, and the surcharge subtracts 6 USD. Regular demand and no surcharge each contribute 0 USD.

Task 1. Identify the outcomes and their probabilities. List every complete outcome in demand–surcharge order and multiply within each pair to calculate its probability.

Task 2. Identify the random variable values and their probabilities. Define M, the net change in profit, with units. Map every outcome to its value of M. List the support. Group outcomes by value to build the PMF, show your calculations, and verify nonnegative masses and a sum of 1.

Task 3. Find a specific probability. Use the PMF to find $P(M = 0)$, the probability that the day's profit equals the baseline. Interpret the result.

Independent practice (continued)

7. Market stall rental credit **Type A**

A vendor is assigned exactly one market location. The standard stall fee is 40 USD. The table gives the fee the vendor actually pays at each location. The rental credit is the difference between the standard fee and the fee paid.

Location	Front gate	Main aisle	Side aisle	Rear row	Outdoor row
Assignment probability	0.12	0.23	0.18	0.27	0.20
Fee paid (USD)	40	40	30	30	20

Task 1. Identify the outcomes and their probabilities. Record every market location outcome and its assignment probability.

Task 2. Identify the random variable values and their probabilities. Define R, the rental credit, with units. Subtract the fee paid from 40 USD to map each location to its credit. List the support. Group outcomes by value to build the PMF, show your calculations, and verify nonnegative masses and a sum of 1.

Task 3. Find a specific probability. Use the PMF to find $P(R = 10)$, the probability that the vendor receives exactly 10 USD in rental credit. Interpret the result.

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8. An emergency restock trip **Type B**

Two stores independently request (R) or do not request (N) an emergency restock. The probabilities of a request are 0.15 for Store 1 and 0.40 for Store 2. If at least one requests a restock, the distributor makes a single trip at a cost of 9 USD plus 3 USD for each requesting store. With no requests, the cost is 0 USD.

Task 1. Identify the outcomes and their probabilities. List every complete request/no-request outcome in store order and multiply within each pair to calculate its probability.

Task 2. Identify the random variable values and their probabilities. Define D, the distributor's restock cost, with units. Map every outcome to its value of D. List the support. Group outcomes by value to build the PMF, show your calculations, and verify nonnegative masses and a sum of 1.

Task 3. Find a specific probability. Use the PMF to find $P(D = 12)$, the probability that the distributor spends exactly 12 USD on the emergency restock. Interpret the result.

Solutions

Complete outcomes Variable values and units Probabilities

3. Courier pay adjustment — Type A

Task 1. Identify the outcomes and their probabilities. Early, on time, slightly late, and very late have probabilities 0.18, 0.47, 0.23, and 0.12, respectively.

Task 2. Identify the random variable values and their probabilities. A measures the change from the courier's base pay for one delivery, in USD. Early and on time each give 2. Slightly late gives 0 and very late gives -3. Add the early and on-time probabilities because these mutually exclusive outcomes both give $A = 2$. The PMF is $p_A(-3) = 0.12$, $p_A(0) = 0.23$, and $p_A(2) = 0.18 + 0.47 = 0.65$. The support is $\{-3, 0, 2\}$ in USD. All masses are nonnegative and $0.12 + 0.23 + 0.65 = 1$.

Task 3. Find a specific probability. From the PMF, $P(A = 2) = p_A(2) = 0.65$. The courier receives 2 USD above base pay with probability 0.65.

4. Delivery fees from two orders — Type B

Task 1. Identify the outcomes and their probabilities. The probabilities of keeping the orders are $1 - 0.20 = 0.80$ and $1 - 0.35 = 0.65$. Multiply within each ordered outcome by independence.

Ordered outcome	KK	KC	CK	CC
Outcome probability	$0.80 \cdot 0.65 = 0.52$	$0.80 \cdot 0.35 = 0.28$	$0.20 \cdot 0.65 = 0.13$	$0.20 \cdot 0.35 = 0.07$

Task 2. Identify the random variable values and their probabilities. F is the total fee revenue in USD. Each kept order contributes 3 USD. The outcomes KK, KC, CK, CC give 6, 3, 3, 0, respectively. Add across the mutually exclusive outcomes KC and CK, which both give $F = 3$. The PMF is $p_F(0) = 0.07$, $p_F(3) = 0.28 + 0.13 = 0.41$, and $p_F(6) = 0.52$. The support is $\{0, 3, 6\}$ in USD. All masses are nonnegative and $0.07 + 0.41 + 0.52 = 1$.

Task 3. Find a specific probability. From the PMF, $P(F = 3) = p_F(3) = 0.41$. The shop earns exactly 3 USD in delivery fees with probability 0.41.

5. Buying a carton of fruit — Type A

Task 1. Identify the outcomes and their probabilities. The orchard, cooperative, wholesaler, and grocery have probabilities 0.28, 0.19, 0.31, and 0.22, respectively.

Task 2. Identify the random variable values and their probabilities. C measures what the stall pays for one carton, in USD. The orchard and wholesaler each give 14. The cooperative gives 18 and the grocery gives 22. Add the orchard and wholesaler probabilities because only one supplier fills the order and both charge 14 USD. The PMF is $p_C(14) = 0.28 + 0.31 = 0.59$, $p_C(18) = 0.19$, and $p_C(22) = 0.22$. The support is $\{14, 18, 22\}$ in USD. All masses are nonnegative and $0.59 + 0.19 + 0.22 = 1$.

Task 3. Find a specific probability. From the PMF, $P(C = 14) = p_C(14) = 0.59$. The stall pays 14 USD for the carton with probability 0.59.

Solutions (continued)

6. Demand and a supplier surcharge — Type B

Task 1. Identify the outcomes and their probabilities. The regular-demand and no-surcharge probabilities are 0.70 and 0.75. Multiply within each ordered outcome by independence.

Ordered outcome	HS	HN	RS	RN
Outcome probability	$0.30 \cdot 0.25 =$ 0.075	$0.30 \cdot 0.75 =$ 0.225	$0.70 \cdot 0.25 =$ 0.175	$0.70 \cdot 0.75 =$ 0.525

Task 2. Identify the random variable values and their probabilities. M is the day's net change in profit from baseline, in USD. Adding the signed contributions gives $6 - 6 =$ 0 for HS, 6 for HN, -6 for RS, and 0 for RN. Add across the mutually exclusive outcomes HS and RN, which both give $M = 0$. The PMF is $p_M(-6) =$ 0.175, $p_M(0) = 0.075 + 0.525 =$ 0.600, and $p_M(6) =$ 0.225. The support is $\{-6, 0, 6\}$ in USD. All masses are nonnegative and $0.175 + 0.600 + 0.225 = 1$.

Task 3. Find a specific probability. From the PMF, $P(M = 0) = p_M(0) =$ 0.600. The shop's profit equals its baseline profit with probability 0.600.

7. Market stall rental credit — Type A

Task 1. Identify the outcomes and their probabilities. Front gate, main aisle, side aisle, rear row, and outdoor row have probabilities 0.12, 0.23, 0.18, 0.27, and 0.20, respectively.

Task 2. Identify the random variable values and their probabilities. R measures the discount from the standard stall fee, in USD. Subtract the fee paid from 40. The front gate and main aisle each give $40 - 40 =$ 0. The side aisle and rear row each give $40 - 30 =$ 10. The outdoor row gives $40 - 20 =$ 20. Add within each group because the vendor receives exactly one location. The PMF is $p_R(0) = 0.12 + 0.23 =$ 0.35, $p_R(10) = 0.18 + 0.27 =$ 0.45, and $p_R(20) =$ 0.20. The support is $\{0, 10, 20\}$ in USD. All masses are nonnegative and $0.35 + 0.45 + 0.20 = 1$.

Task 3. Find a specific probability. From the PMF, $P(R = 10) = p_R(10) =$ 0.45. The vendor receives a 10 USD rental credit with probability 0.45.

8. An emergency restock trip — Type B

Task 1. Identify the outcomes and their probabilities. The no-request probabilities are 0.85 and 0.60. Multiply within each ordered outcome by independence.

Ordered outcome	RR	RN	NR	NN
Outcome probability	$0.15 \cdot 0.40 =$ 0.06	$0.15 \cdot 0.60 =$ 0.09	$0.85 \cdot 0.40 =$ 0.34	$0.85 \cdot 0.60 =$ 0.51

Task 2. Identify the random variable values and their probabilities. D is the distributor's emergency restock cost in USD. For RR, the cost is $9 + 2 \cdot 3 =$ 15. For RN or NR, it is $9 + 3 =$ 12. For NN, it is 0. Add across the mutually exclusive outcomes RN and NR, which both give $D = 12$. The PMF is $p_D(0) =$ 0.51, $p_D(12) = 0.09 + 0.34 =$ 0.43, and $p_D(15) =$ 0.06. The support is $\{0, 12, 15\}$ in USD. All masses are nonnegative and $0.51 + 0.43 + 0.06 = 1$.

Task 3. Find a specific probability. From the PMF, $P(D = 12) = p_D(12) =$ 0.43. The distributor spends exactly 12 USD on the restock trip with probability 0.43.